

Visual-Language Decision System through Integration of Foundation Models for Service Robot Navigation

Peiyuan Zhu¹, Lotfi El Hafi^{1,*}, and Tadahiyo Taniguchi¹

Abstract—This study aims to build a system that bridges the gap between robotics and environmental understanding by integrating various foundation models. While current visual-language models (VLMs) and large language models (LLMs) have demonstrated robust capabilities in image recognition and language comprehension, challenges remain in integrating them into practical robotic applications. Therefore, we propose a visual-language decision (VLD) system that allows a robot to autonomously analyze its surroundings using three VLMs (CLIP, OFA, and PaddleOCR) to generate semantic information. This information is further processed using the GPT-3 LLM, which allows the robot to make judgments during autonomous navigation. The contribution is twofold: 1) We show that integrating CLIP, OFA, and PaddleOCR into a robotic system can generate task-critical information in unexplored environments; 2) We explore how to effectively use GPT-3 to match the results generated by specific VLMs and make navigation decisions based on environmental information. We also implement a photorealistic training environment using Isaac Sim to test and validate the proposed VLD system in simulation. Finally, we demonstrate VLD-based real-world navigation in an unexplored environment using a TurtleBot3 robot equipped with a lidar and an RGB camera.

I. INTRODUCTION

We increasingly rely on robots to perform service tasks in our daily lives, but they often struggle with unexpected changes in their environments. While recent breakthroughs in visual-language models (VLMs) and large language models (LLMs), *i.e.*, foundation models, have demonstrated robust capabilities in image recognition and language comprehension, challenges remain in integrating them in practical robotic applications.

Therefore, this study aims to build a system that bridges the gap between service robotics and environmental understanding through the integration of various foundation models. Indeed, the system integration of VLMs and LLMs could enhance service robot operations. Examples include translating environmental sensor data into natural language, giving robots the ability to understand users' natural language, planning actions based on natural language instructions or descriptions, or even turning users' natural language into machine-readable code to execute complex robot behaviors.

In this regard, our study focuses on enhancing the decision-making process of service robots during au-

This work was supported by the Japan Science and Technology Agency (JST), Moonshot Research & Development Program, Grant Number JPMJMS2011.

¹Peiyuan Zhu, Lotfi El Hafi, and Tadahiyo Taniguchi are with Ritsumeikan University; 1-1-1 Noji-Higashi, Kusatsu, Shiga 525-8577, Japan. {zhu.peiyuan, lotfi.elhafi, taniguchi}@em.ci.ritsumei.ac.jp

*Corresponding author.

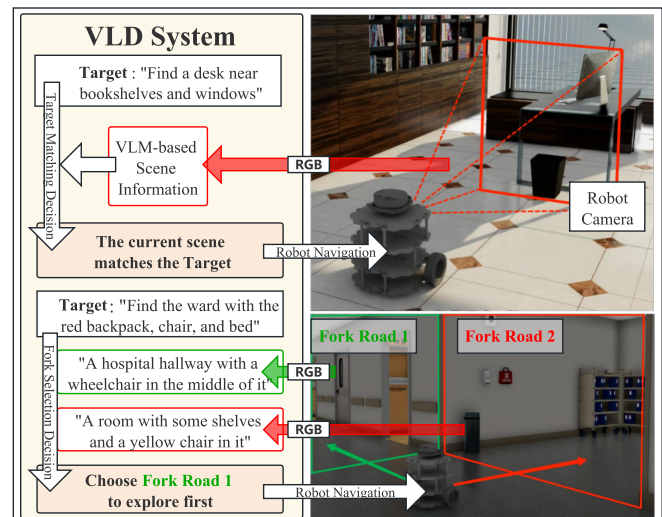


Fig. 1. Overview of the proposed visual-language decision (VLD) system: 1) the VLD system allows a robot to utilize the CLIP, OFA, and PaddleOCR visual-language models (VLMs) to gather environmental information, and 2) the VLD system matches this information with the instructions provided by users using the GPT-3 large language model (LLM) during navigation in unexplored environments. The system integration was tested and validated in both virtual simulations (Nvidia Isaac Sim) and real-world experiments (Robotis TurtleBot3).

tonomous navigation through the system integration of VLMs and LLMs. More specifically, we propose a visual-language decision (VLD) system that allows a robot to generate semantic information about its surroundings using three VLMs (CLIP, OFA, PaddleOCR) and to make judgments during autonomous navigation with an LLM (GPT-3). Consequently, our proposed VLD system enables the robot to collect and interpret environmental features and decide its next motion based on users' instructions, as illustrated by 1.

The primary contribution of our proposed VLD system is twofold:

- 1) We show that integrating VLMs (CLIP, OFA, and PaddleOCR) into a robotic system can generate task-critical information when a robot faces unexplored environments.
- 2) We explore how to effectively use an LLM (GPT-3) to judge and match the results generated by specific VLMs, subsequently making navigation decisions based on environmental information.

Additionally, we implemented a photorealistic training environment using Nvidia Isaac Sim to test and validate the proposed VLD system in simulation. More importantly, we validated the proposed VLD system in a real-world naviga-

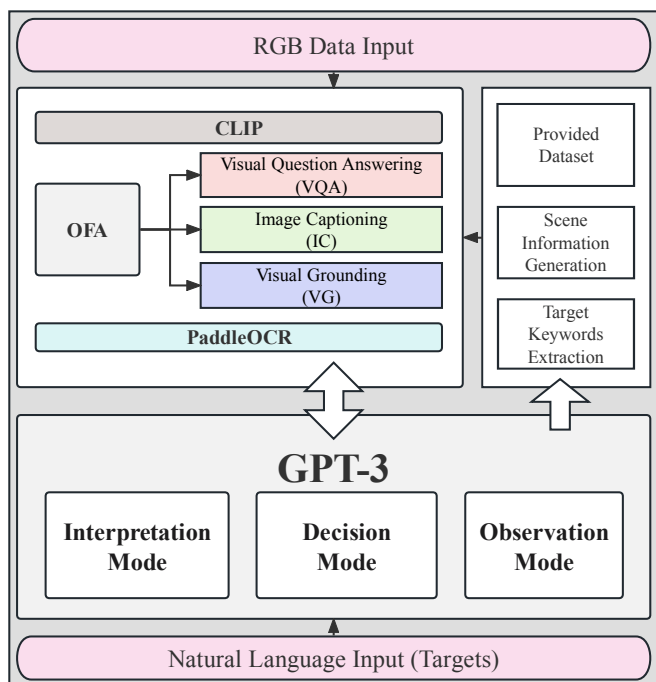


Fig. 2. Overview of the integration of the proposed VLD system. Specific data exchanges and judgment functionality between each software module are further described in Fig. 3.

tion experiment using a Robotis TurtleBot3 robot equipped with a lidar and an RGB camera, demonstrating VLD-based navigation in an unexplored environment.

The remainder of this paper is structured as follows. Section II introduces the foundation models used in the proposed system. Section III details the integration of the proposed system. Section IV describes the experimental setups used to validate the proposed system. Section V summarizes the experimental results obtained with the proposed system. Section VI analyzes the advantages and limitations of the proposed system. Finally, Section VII concludes this study with avenues for future improvements.

II. RELATED WORK

On the one hand, the proposed VLD system relies on the following VLMs to gather environmental information: CLIP, OFA, and PaddleOCR. CLIP [1], from OpenAI, can understand both images and text, excelling in tasks such as zero-shot image classification and image captioning. OFA [2], from Alibaba Damo Academy, presents various visual capabilities, including visual question answering (OFA-VQA), image captioning (OFA-IC), and visual grounding (OFA-VG). PaddleOCR [3], from Baidu, focuses on optical character recognition (OCR) in multiple languages and supports continuous real-time text detection and scene description. However, outside of the convenience of performance evaluations using generic computer vision datasets, the practical application of these advanced visual capabilities remains difficult in the context of service robotics.

On the other hand, LLMs such as GPT-3 [4] have demonstrated the ability to control robots by translating natural

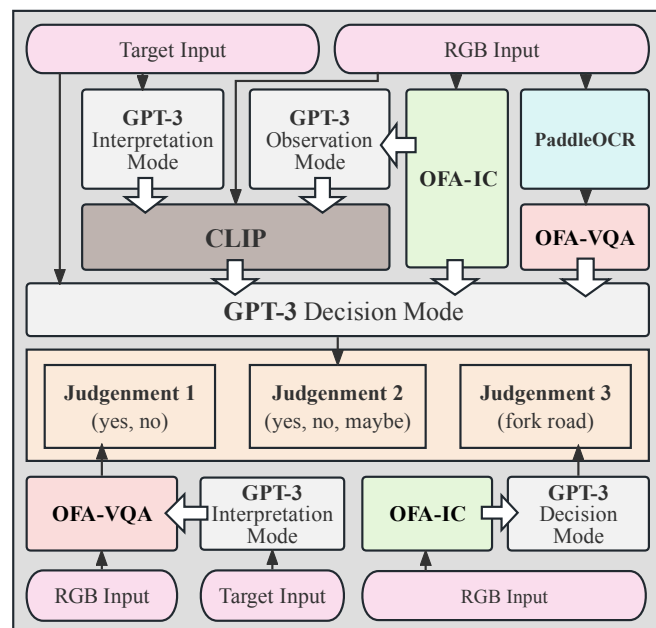


Fig. 3. Judgment functionality and data exchanges between each software module of the proposed VLD system. The decision flow in the proposed VLD system is generated through several judgment modules that match the scene with the instructed target and select the path at a navigation fork. On the one hand, Judgment 1 and Judgment 2 generate further actions to refine decisions following specific output combinations of “yes”, “maybe”, or “no”, as described in Table II. On the other hand, Judgment 3 directly selects the most appropriate path at a navigation fork.

language instructions into machine-readable commands to execute complex robotic behaviors. This particular application of LLM was notably described in an early study [5] by Microsoft that employed GPT-3 in various robotic tasks, including service and navigation. However, many similar studies have often focused on a single LLM and have not yet considered the integration of several foundational models. Therefore, in our study, we leverage GPT-3’s natural language processing abilities to interpret the outputs from the three VLMs described previously. This enables robots to better understand their surrounding environment, including specific local observations, and follow users’ instructions more accurately.

Similarly to our aim, OFA-OCR [6] was recently proposed. OFA-OCR combined pre-trained VLMs, such as CLIP and OFA, to enhance robotic recognition behaviors, including object, shape, and color recognition, to generate detailed environmental information. However, the integration of the OFA-OCR system was basic, and although the method could provide scene data, processing this information for robot decision-making remained challenging. Therefore, in our study, we test and validate our proposed VLD system’s integration using photorealistic simulations and a real robot.

III. PROPOSED SYSTEM

An overview of the integration of the proposed VLD system is given in Fig. 2. Specific data exchanges and judgment functionalities between each software module of the proposed VLD system are described in Fig. 3. Additionally,

TABLE I
GPT-3 PROMPT TEMPLATES

Function	Template	Output
Target Dataset	Separate the nouns that refer to specific scenes but not the specific objects/scenes in the following words, or return "none" if none. Do not describe, do not add "and": "{target}"	(scenes/objects)
VQA-Generation	Analyze the following sentence. If there is a sentence describing the positional relationship between objects (for example, "a banana is on the left side of an apple"), then generate a question sentence asking about the positional relationship between objects (for example, "is a banana on the left side of an apple?"). If not (for example, "there are chairs in the room"), output interrogative sentences such as "is there a chair in this room?". Do not make other descriptions or explanations: "{target}"	(question)
GPT-Matching	I am looking for the target "{target}". My current environment is described as: "{description}". Does this target exist in the given scenario? Answer with "yes" or "maybe" or "no".	yes, maybe, no
Fork Selection	I am at a crossroads with three paths labeled as "r1", "r2", and "r3". Their descriptions are as follows: "r1: '{description1}'","r2: '{description2}'","r3: '{description3}'". I am looking for target "{target}". Based on these descriptions, which path is most likely to lead me to the target? Only answer "r1", "r2" or "r3". Do not describe.	r1, r2, r3
IC Dataset	In the environment called "{description}", list the possible objects that might exist. Provide the output as a comma-separated list, with each description being less than 50 words and without using the conjunction "and". Do not describe.	(scenes/objects)

TABLE II
JUDGMENT CRITERIA FOR TARGET MATCHING AND FORK SELECTION

Judgement 1	Judgement 2	Decision
yes	yes/maybe	Confirm the target position through OFA-VG, instruct the robot to stop, and complete the navigation task.
yes	no	Input target into OFA-VG to get a bounding box, crop the bounding box, and input it again as RGB data.
no	yes	
no	maybe/no	Determine that the scene does not match the target, and select the specified fork according to the result of Judgement 3 to continue exploring the environment.

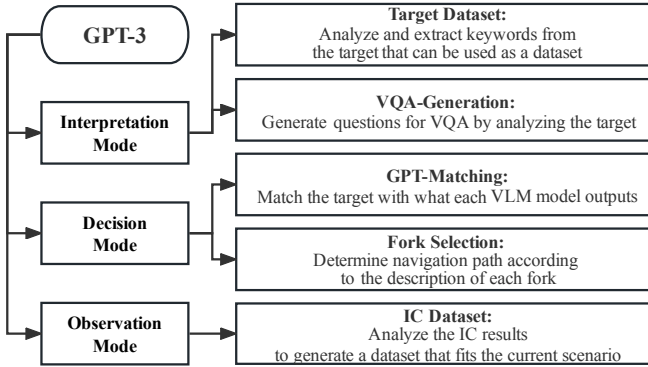


Fig. 4. The proposed VLD system employs GPT-3 in three distinct modes with several functions: interpretation mode, observation mode, and decision mode. The specific implementation of each function is detailed in Table I.

Table II describes the two main decision-making processes of the proposed VLD system: One matches the scene with the target, and the other selects the path when encountering a navigation fork.

A. Visual-Language Models

The proposed VLD system generates semantic information about the robot's surrounding environment using three VLMs: CLIP [1], OFA [2], and PaddleOCR [3].

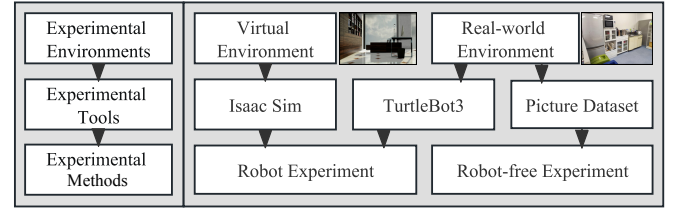


Fig. 5. Overview of the experimental setups used to test and validate the proposed VLD system.

1) *CLIP*: In the VLD system, CLIP mainly provides additional cues for the target-matching decision function of the system. Because the response time of other VLMs is too long and the length of the generated text is limited, important keywords can be missed. On the other hand, CLIP can be invoked several times quickly, improving the matching accuracy in the decision-making process of the proposed VLD system.

More specifically, the CLIP model matches the image with the dataset multiple times, selects the element with the highest score after each match, and continues to match with the remaining dataset until the highest score is less than 20%. These matched elements are then used as cues for the target-matching decision function.

2) *OFA*: The proposed VLD system utilizes three core functions of the OFA model: visual question answering (OFA-VQA), image captioning (OFA-IC), and visual grounding (OFA-VG).

- Visual question answering (OFA-VQA): Binary VQA (BVQA) enables questions like "is {...}?" or "does {...}?" to determine the presence of specific properties in an image. Matching VQA (MVQA) enables questions such as "what type of {...}?" to extract specific details from the image. Using both BVQA and MVQA offers a comprehensive understanding of the scene, facilitating task completion by the robot.
- Image captioning (OFA-IC): OFA-IC generates detailed descriptions of images, capturing the main objects,

TABLE III
EXAMPLES OF MATCHING RESULTS IN EXPERIMENT 1 (KITCHEN, CORRIDOR, LABORATORY)

	kitchen	corridor	laboratory
Environment Pictures			
Target	“Some coffee machines in the kitchen”	“A room numbered 123”	“A bustling laboratory”
Target Matching Decision	yes	yes	yes
Target	“Microwave located to the right of the refrigerator”	“A room numbered 122”	“A laboratory that has not been cleaned in a long time”
Target Matching Decision	yes	no	no
Target	“Microwave located to the right of the refrigerator in a shop”	“Microwave located to the right of the refrigerator”	“A bright laboratory”
Target Matching Decision	no	no	yes

TABLE IV
EXAMPLE OF DETAILED RESULTS IN EXPERIMENT 1 (KITCHEN)

Target	“Some coffee machines in the kitchen”	“Microwave located to the right of the refrigerator”	“Microwave located to the right of the refrigerator in a shop”
VQA-Generation	“Are there coffee machines in the kitchen?”	“Is the microwave located to the right of the refrigerator?”	“Is the microwave located to the right of the refrigerator in a shop?”
Target Dataset	coffee machines, kitchen	microwave, refrigerator	microwave, refrigerator, shop
Image Captioning	“A kitchen with a refrigerator microwave and other appliances”		
IC Dataset	refrigerator, microwave, oven, stove, sink, dishwasher, toaster, blender, coffee maker, kettle, cutting board, knife set, pots, pans, ...		
CLIP	kitchen		
OFA-VQA	yes	yes	no
GPT-Matching	maybe	maybe	no
Target Matching Decision	yes	yes	no

context, and potential actions. This aids the robot in its decision-making process.

- Visual grounding (OFA-VG): OFA-VG can effectively replace conventional object recognition models, such as YOLO [7], which require pre-training to identify regions of interest (ROIs), *i.e.*, bounding boxes in a given image, that correspond to a sentence. The ROIs output by OFA-VG can be further processed using OFA-VQA for enhanced object descriptions.

3) *PaddleOCR*: Our preliminary experiments revealed that VLMs like CLIP and OFA had difficulties with tasks involving specific text information, such as floor maps or

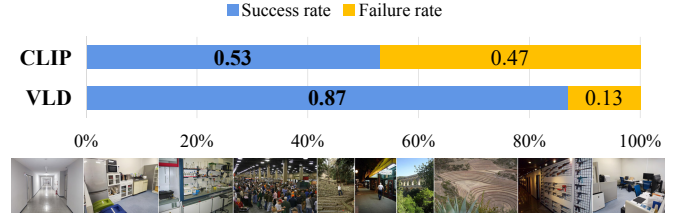


Fig. 6. Comparison of average success rates in target matching between the proposed VLD system and the CLIP baseline for 100 pictures of real-world scenes repeated five times.

wall signs. Specifically, OFA faced challenges recognizing objects containing specific text, while CLIP struggled with tasks involving closely related numbers.

To address these issues, we integrated PaddleOCR into the proposed VLD system to perform accurate optical character recognition (OCR) in the robot’s surrounding environment. Furthermore, we combined PaddleOCR with the VQA capabilities of OFA to discern on which objects the text is written, subsequently achieving higher accuracy than when using OFA-VQA alone.

B. Large Language Model

The proposed VLD system leverages GPT-3 [4], an LLM, for natural language processing (NLP) which is crucial for interpreting the results of the VLMs. Indeed, GPT-3 provides in-depth semantic understandings, enhancing the robot’s perception of the environment and its accuracy in executing commands. Specifically, we employ GPT-3 in three distinct modes with several functions, as described by Fig. 4 and Table 4: interpretation mode, observation mode, and decision mode.

1) Natural Language Processing (Interpretation Mode):

The proposed VLD system employs GPT-3 to process natural language data by 1) extracting essential keywords from the

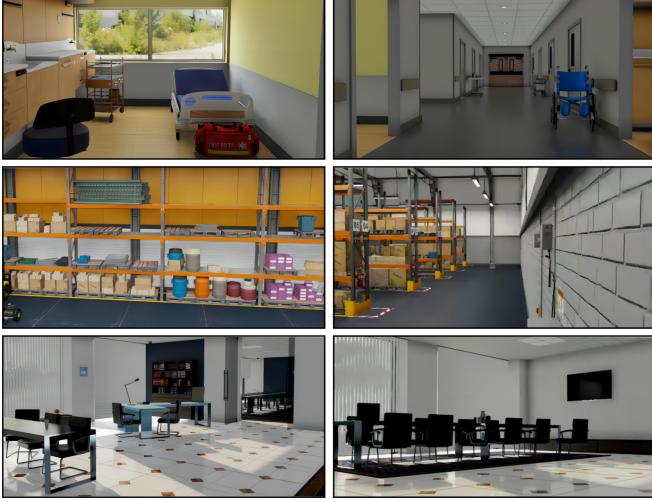


Fig. 7. Virtual environments used in Experiment 2 and rendered by Nvidia Isaac Sim. Top-down: hospital, warehouse, and office.

user’s targets, such as scene or object names, and 2) generating questions for VQA functions in the system based on these targets. By querying OpenAI’s API, we can effectively utilize the `gpt-3.5-turbo` model to achieve real-time performance for these two purposes.

2) *Local Dataset Generation (Observation Mode)*: Relying solely on pre-made general datasets is inadequate when facing a new environment because they do not include specific local knowledge or observations. The proposed VLD system employs GPT-3 to address this as follows:

- We prepare small datasets containing everyday objects or scenes.
- We use GPT-3 to process the user’s targets, deriving nouns related to objects or scenes to create a new dataset.
- We feed GPT-3 with image captions obtained from OFA-IC, enabling it to predict potential objects or scenes.
- We combine these datasets to provide a comprehensive yet efficient dataset tailored to the current environment.

This approach enhances the robot’s adaptability in different environments, as it does not require preparing a specific large dataset in advance.

3) *Scene Text Matching (Decision Mode)*: The proposed VLD system employs GPT-3 to compare generated scene descriptions to human-defined objectives. Despite utilizing several VLMs for detailed textual environment descriptions, identifying exact matches to a target is challenging. By inputting both the targets and scene descriptions into GPT-3, we prompt it to gauge the alignment, using outcomes like “yes”, “maybe”, or “no” to determine decision-making.

IV. EXPERIMENTS

Fig. 5 gives an overview of the experimental setups. We used two experimental environments with two experimental methods and conducted a total of three experiments.



Fig. 8. Apartment mock-up with various domestic items used in Experiment 3 to validate the proposed VLD system in a real-world navigation task using a Robotis TurtleBot3 robot.

A. Robot-Free Experiment with Picture Datasets (Experiment 1)

Experiment 1 tests the proposed VLD system’s accuracy in matching real-world pictures with designated targets. The system performance is evaluated on individual images, ensuring no prior information is used. Using such pictures, we can test whether the system can handle complex environments. The CLIP model serves as a baseline for quantitatively comparing success rates in target matching.

Specifically, we use a dataset of 100 real-world pictures. Each picture is given a specific target, and then matched using the CLIP model and the VLD system. The targets are diverse and include, for example: “A man was standing on the steps”, “Many toy guns on the wall”, or “Microwave located to the right of the refrigerator”. The experiment is conducted five times, and the average success rates in target matching are recorded.

B. Robot Experiment in Virtual Environments (Experiment 2)

Experiment 2 is the primary focus of the study, designed to test whether the proposed VLD system can perform in various complex environments. For this purpose, we utilize Nvidia Isaac Sim to render photorealistic environments and conduct virtual simulations. Specifically, we used three indoor environments as illustrated in Fig. 7: hospital, warehouse, and office. These environments present diverse visual features and different navigation challenges.

In addition, Nvidia Isaac Sim publishes the virtual sensor data via ROS messages, *i.e.*, topics, similar to the real hardware robot. Therefore, the proposed VLD system subscribes and publishes to these ROS topics for controlling and navigating the robot in simulation.

Similarly to Experiment 1, the CLIP model serves as a baseline for quantitatively comparing success rates in target matching. Five targets are specified for each virtual environment, and then matched using the CLIP model and the VLD system, for a total of 15 different scenarios. For example, hospital: “Find a hospital ward with the red backpack, chair, and bed”, “A hallway with a wheelchair”, “A room without a bed”, “Red backpack on wooden floor”,



Fig. 9. Example of results in the hospital virtual environment of Experiment 2. At the first and second navigation forks, the VLD system recognizes that the scene mismatches the target in both fork directions, and selects Fork 1 for further exploration. At the third navigation fork, the VLD system recognizes that the scene in the direction of Fork 2 matches the target, completing the mission.

“A bright room with a wheelbarrow”; warehouse: “Find some road cones in the warehouse”, ... ; office: “Find a desk near bookshelves and windows”, The experiment is conducted five times, and the average success rates in target matching are recorded.

C. Robot Experiment in a Real-World Environment (Experiment 3)

Experiment 3 aims to validate the proposed VLD system in a real-world navigation task using a Robotis TurtleBot3 robot equipped with a lidar and an RGB camera. For this purpose, we constructed an apartment mock-up using various domestic items such as a TV, tables, chairs, and tableware, as illustrated by Fig. 8. The interface of the TurtleBot3 hardware with ROS was similar to the configuration used in Experiment 2.

V. RESULTS

This section briefly summarizes the experimental results of each experiment. For each experiment, detailed results of one of the navigation scenarios are given.

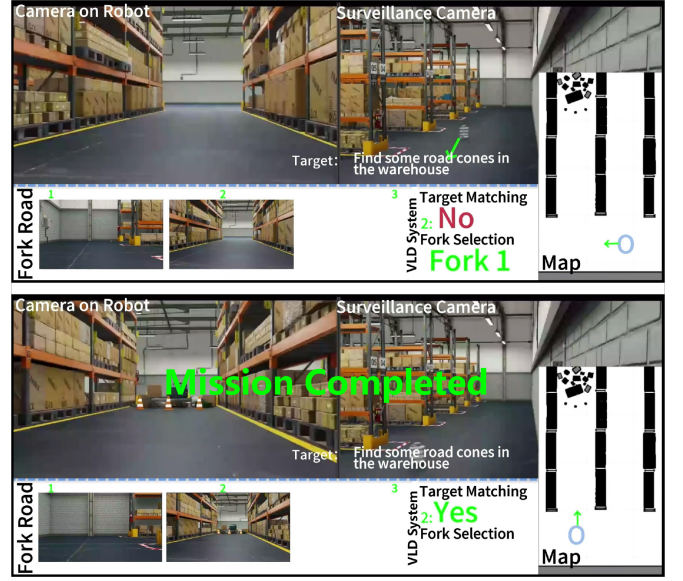


Fig. 10. Example of results in the warehouse virtual environment of Experiment 2. At the first navigation fork, the VLD system recognizes that the scene mismatches the target in both fork directions, and selects Fork 1 for further exploration. At the second navigation fork, the VLD system recognizes that the scene in the direction of Fork 2 matches the target, completing the mission.

A. Robot-Free Experiment with Picture Datasets (Result 1)

Table III presents results obtained in the robot-free experiment for three sample pictures from the dataset: kitchen, corridor, and laboratory. In addition, the detailed results for the kitchen scenario are given in Table IV. Finally, the comparison of average success rates in target matching between the VLD system and the CLIP baseline for 100 pictures is shown in Fig. 6.

B. Robot Experiment in Virtual Environments (Result 2)

Examples of the navigation decision results in each virtual environment (hospital, warehouse, office) are shown in Fig. 9, Fig. 10, and Fig. 11. In addition, the detailed results of one scenario in the hospital environment are given in Table V. Finally, the comparison of average success rates in target matching between the VLD system and the CLIP baseline for 15 scenarios is shown in Fig. 13.

C. Robot Experiment in a Real-World Environment (Result 3)

An example of the navigation decision results in the real-world apartment environment is shown in Fig. 12. The detailed results of one scenario in the environment are given in Table VI.

VI. DISCUSSION

Result 1 shows that, by combining various VLMs with an LLM, the proposed VLD system could generate questions for VQA in the displayed scene and prepare the dataset according to the instructions. Therefore, GPT-3 could effectively generate scenario-specific datasets for the CLIP model to provide basic data for the system's decision-making module.



Fig. 11. Example of results in the office virtual environment of Experiment 2. At the first navigation fork, the VLD system recognizes that the scene mismatches the target in all three fork directions, and selects Fork 3 for further exploration. At the second navigation fork, the VLD system recognizes that the scene in the direction of Fork 2 matches the target, completing the mission.

Moreover, the proposed VLD system could better match the targets to achieve higher success rates compared to the CLIP baseline.

Result 2 shows that, in photorealistic simulation environments, the robot could navigate and recognize paths autonomously by assessing environmental cues. In other words, the robot could successfully recognize objects and scenes to make navigation decisions through the proposed VLD system. Furthermore, the experiments also showed that the proposed VLD system had higher accuracy in target matching than the CLIP baseline.

Result 3 shows that, in a real-world experiment, the robot could find the target scenes in the surrounding environment through the proposed VLD system. However, there were often mismatches due to the camera orientation and image resolution.

Analyzing the above results, the VLD system revealed some advantages and limitations.

- **Advantages:** The proposed VLD system utilizes OpenAI's GPT-3 model, allowing efficient natural language processing with minimal resource use. This enhances the overall flexibility and scalability of the system. Furthermore, by integrating CLIP, OFA, and PaddleOCR VLMs, the proposed VLD system achieves superior environmental understanding, including specific local observations, compared to systems that use a single foundation model. This supports our initial experimental hypothesis.
- **Limitations:** The proposed VLD system struggles to utilize environmental information fully and cannot deduce object locations from, for example, floor maps

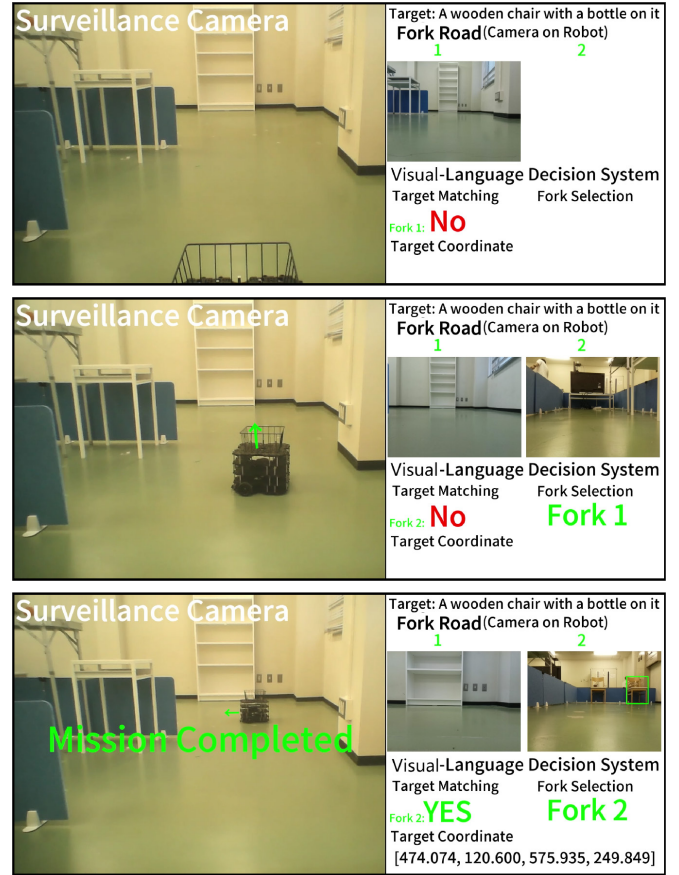


Fig. 12. Example of results in the apartment real-world environment of Experiment 3. At the first navigation fork, the VLD system recognizes that the scene mismatches the target in both fork directions, and selects Fork 1 for further exploration. At the second navigation fork, the VLD system recognizes that the scene mismatches the target in both fork directions, and selects Fork 1 for further exploration. At the third navigation fork, the VLD system recognizes that the scene in the direction of Fork 2 matches the target and marks its location, completing the mission.

or wall signs. Abstract and ambiguous targets are not well understood and recognized by the system. Real-world tests also revealed that, due to the Turtlebot3 robot's limited viewing angle and camera quality, the proposed VLD system could not deal with noisy image data. Additionally, the absence of a mapping function sometimes results in repeated searches in the same locations, increasing computational cost and completion time.

We found that by adjusting the parameters to make VLMs output more text, *i.e.*, increasing the outputs of VQA and IC, the success rate can be improved, but at a reduced overall system efficiency.

VII. CONCLUSION

In this study, we integrated a visual-language decision (VLD) system that combines three VLMs (CLIP, OFA, PaddleOCR) with an LLM (GPT-3) to enhance the decision-making capabilities of service robots during navigation. More specifically, we demonstrated in both photorealistic simulations and real-world experiments that, by leveraging multiple

TABLE V
EXAMPLE OF DETAILED RESULTS IN EXPERIMENT 2 (HOSPITAL)

	Fork Road 1	Fork Road 2
Fork Road Pictures		
Target	"Find a hospital ward with the red backpack, chair, and bed"	
VQA-Generation	"Is there a hospital ward with a red backpack, chair, and bed?"	
Target Dataset	hospital ward, red backpack, chair, hospital bed	
Image Captioning	"A hospital hallway with a wheelchair and a bike in the middle of it"	"A room with a chair and a desk in front of a window"
IC Dataset	hospital hallway, wheelchair, medical equipment, doors, walls, floors, lights, signs, ...	chair, table, desk, bookshelf, lamp, window, door, computer, plant, clock, whiteboard, book, pen, notebook, cabinet, ...
CLIP	hallway (score: 0.18)	office (score: 0.33)
OFA-VQA	no	no
GPT-Matching	no	no
Target Matching Decision	no	no
Fork Selection Decision	Fork Road 1	

TABLE VI
EXAMPLE OF DETAILED RESULTS IN EXPERIMENT 3 (APARTMENT)

	Fork Road 1	Fork Road 2
Fork Road Pictures		
Target	"A wooden chair with a bottle on it"	
VQA Generation	"Is there a bottle on the wooden chair?"	
Target Dataset	wooden chair, bottle	
Image Captioning	"A room with a shelf on the wall and a gray floor"	"A large room with blue walls and a floor with white cones"
IC Dataset	shelf, gray floor, chair, table, lamp, rug, window, book, picture, clock, plant, mirror, couch, ottoman, cupboard, ...	
CLIP	gray floor	gray floor
OFA-VQA	no	yes
GPT-Matching	maybe	maybe
Target Matching Decision	no	yes
Fork Selection Decision	Fork Road 2	
Target Location		[474.074, 120.600, 575.935, 249.849]

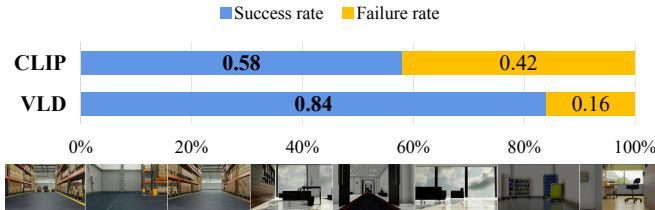


Fig. 13. Comparison of average success rates in target matching between the proposed VLD system and the CLIP baseline for five targets in each virtual environment (hospital, warehouse, office), for a total of 15 scenarios repeated five times.

VLMs to interpret scenes and channeling this information through an LLM for further processing, the proposed VLD system can allow robots to autonomously navigate according to users' instructions in unexplored environments without prior knowledge.

However, the VLD System is also prone to errors when the target objects are too small or abstract. This requires finer image segmentation and a better semantic understanding. Therefore, in future improvements, we plan to integrate VLMs [8] into the system to map the unexplored environment. We also aim to employ VLN [9] to enhance navigation further. Additionally, we want to incorporate LSeg [10] to better segment objects in the scene. This verification will be conducted as a contribution to the JST Moonshot R&D Program, Goal 1 [11], [12].

REFERENCES

- [1] A. Radford, *et al.*, "Learning Transferable Visual Models from Natural Language Supervision," in *Proceedings of the 38th International*

- Conference on Machine Learning (ICML 2021)*, vol. 139, (Virtual), July 2021, pp. 8748–8763.
- [2] P. Wang, *et al.*, "OFA: Unifying Architectures, Tasks, and Modalities Through a Simple Sequence-to-Sequence Learning Framework," in *Proceedings of the 39th International Conference on Machine Learning (ICML 2022)*, vol. 162, Baltimore, United States, July 2022, pp. 23 318–23 340.
- [3] Y. Du, *et al.*, "PP-OCR: A Practical Ultra Lightweight OCR System," Sept. 2020. [Online]. Available: <https://doi.org/10.48550/arXiv.2009.09941>
- [4] T. Brown, *et al.*, "Language Models are Few-Shot Learners," in *Proceedings of 2020 Conference on Neural Information Processing Systems (NeurIPS 2020)*, vol. 33, (Virtual), Dec. 2020, pp. 1877–1901.
- [5] S. Vemprala, *et al.*, "ChatGPT for Robotics: Design Principles and Model Abilities," Feb. 2023. [Online]. Available: <https://doi.org/10.48550/arXiv.2306.17582>
- [6] J. Lin, *et al.*, "Transferring General Multimodal Pretrained Models to Text Recognition," Dec. 2022. [Online]. Available: <https://doi.org/10.48550/arXiv.2212.09297>
- [7] J. Redmon, *et al.*, "You Only Look Once: Unified, Real-Time Object Detection," in *Proceedings of 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2016)*, Las Vegas, United States, June 2016, pp. 779–788.
- [8] C. Huang, *et al.*, "Visual Language Maps for Robot Navigation," Oct. 2022. [Online]. Available: <https://doi.org/10.48550/arXiv.2210.05714>
- [9] P. Anderson, *et al.*, "Vision-and-Language Navigation: Interpreting Visually-Grounded Navigation Instructions in Real Environments," in *Proceedings of 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2018)*, Salt Lake City, United States, June 2018, pp. 3674–3683.
- [10] B. Li, *et al.*, "Language-driven Semantic Segmentation," in *Proceedings of 10th International Conference on Learning Representations (ICLR 2022)*, (Virtual), Apr. 2022.
- [11] H. Ishiguro, "The Realisation of an Avatar-Symbiotic Society Where Everyone can Perform Active Roles without Constraint," *RSJ Advanced Robotics (AR)*, vol. 35, no. 11, pp. 650–656, June 2021.
- [12] T. Taniguchi, *et al.*, "Semiotically Adaptive Cognition: Toward the Realization of Remotely-Operated Service Robots for the New Normal Symbiotic Society," *RSJ Advanced Robotics (AR)*, vol. 35, no. 11, pp. 664–674, June 2021.